

Software Requirements Specification for Software Engineering: subtitle describing software

Team 11, technically functional

Matthew Huynh

Cieran Diebolt

Vaisnavi Shanthamoorthy

Maham Siddiqui

Eman Ashraf

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Revision History

Date	Version	Notes
Date 1	1.0	Notes
Date 2	1.1	Notes

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1.2 Goals of the Project

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2.7 User Participation

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2.8 Maintenance Users and Service Technicians

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3 Mandated Constraints

3.1 Solution Constraints

The system shall provide users access to their account and previous exercise information, provided a valid username and password.

Justification: User data stored in the user database is important to the account, but should be safely guarded without validation.

The system shall provide users with an overview of their exercises and data specific to individual exercises.

Justification: Users should be able to look back on their progress and see data on their exercises over time.

The system shall provide users the ability to upload video-feeds of them performing exercises, and return recommendations on their form.

Justification: The primary function of the system, users must be able to get form recommendations to assist in proper recovery and boost confidence in correct form. The system must be able to accept the video as input and be capable of returning relevant recommendations to the user.

3.2 Implementation Environment of the Current System

- **Back End** (Processing): Source code written in Python.
- **Front End** (GUI): Source code written in platform specific application studio (EG. Android Studio).
- **OpenCV:** Library used to analyze user video-streams for required data.

- **Database:** Remotely accessible storage for user data.

3.3 Partner or Collaborative Applications

Physiotherapist scheduling, management, and patient-oriented softwares could be used in tandem, however the system is not designed with collaborative applications in mind.

3.4 Off-the-Shelf Software

- **Database** software is previously implemented and simply interfaced by the system.
- **OpenCV** software is unmodified and only interfaced by the system.

3.5 Anticipated Workplace Environment

- Physiotherapist patients in their residences or private spaces.
- Physiotherapists in their offices or assigned exercise spaces.

3.6 Schedule Constraints

As per the [Development Plan](#):

- Proof of Concept must be developed by November 17th, 2025.
 - Demonstrations: November 17th - 28th.
- Revision 0 must be developed by February 2nd, 2026.
 - Demonstrations: February 2nd - 13th.
- Revision 1 must be developed by March 23rd, 2026.
 - Demonstrations: March 23rd - 29th.

3.7 Budget Constraints

The development of the system must remain less than \$125 or the team begins paying out of pocket for development tools. Ideally the implementation remains free and open to use.

3.8 Enterprise Constraints

The system will attempt to mimic standards set by professional physiotherapists across Canada. Due to the nature of health care and physiotherapy market within Canada, the system will not provide medical care or medical assistance, simply wellbeing advice similar to that of a gym companion application.

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Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. How many of your requirements were inspired by speaking to your client(s) or their proxies (e.g. your peers, stakeholders, potential users)?
4. Which of the courses you have taken, or are currently taking, will help your team to be successful with your capstone project.
5. What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project? Examples of possible knowledge to acquire include domain specific knowledge from the domain of your application, or software engineering knowledge, mechatronics knowledge or computer science knowledge. Skills may be related to technology, or writing, or presentation, or team management, etc. You should look to identify at least one item for each team member.
6. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?